

JINAN, China

WMO: 548230

Lat: 36.6019N

Lon: 117.0081E

Elev: 58

StdP: 100.63

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.4	-6.6	-23.2	0.5	-3.1	-21.2	0.6	-2.7	8.7	3.7	7.7	3.0	2.3	110	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	35.2	23.3	34.0	23.5	32.7	23.2	27.1	32.0	26.5	31.1	25.8	30.2	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.2	30.0	25.1	20.4	29.6	24.5	19.6	28.9	86.1	32.1	83.3	31.2	80.3	30.4	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.4	6.4	DB	-11.1	37.6	2.2	1.6	-12.7	38.7	-14.0	39.7	-15.2	40.5	-16.8	41.7	(4)
(5)			WB	-12.7	28.3	2.0	0.8	-14.1	28.8	-15.2	29.3	-16.3	29.7	-17.7	30.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	0.1	3.4	9.6	16.5	22.2	26.5	27.8	26.3	22.3	16.6	8.7	2.1	(6)
(7)		DBStd	10.42	3.66	4.41	4.90	4.63	3.85	3.13	2.61	2.71	3.17	3.99	4.76	3.78	(7)
(8)		HDD10.0	893	307	187	66	5	0	0	0	0	0	3	79	246	(8)
(9)		HDD18.3	2234	565	417	272	89	11	1	0	0	6	81	290	504	(9)
(10)		CDD10.0	2803	0	3	54	199	378	495	551	505	369	207	41	1	(10)
(11)		CDD18.3	1103	0	0	3	32	131	245	293	247	125	27	1	0	(11)
(12)		CDH23.3	10858	0	0	12	214	1139	2748	3377	2406	852	109	1	0	(12)
(13)		CDH26.7	4395	0	0	1	46	379	1255	1543	929	228	13	0	0	(13)
(14)	Wind	WSAvg	2.9	2.6	3.0	3.5	3.7	3.4	3.1	2.6	2.4	2.5	2.7	2.9	2.6	(14)
(15)	Precipitation	PrecAvg	763	6	14	14	32	72	95	211	186	67	32	26	8	(15)
(16)		PrecMax	1110	38	36	44	113	162	230	408	418	269	123	116	34	(16)
(17)		PrecMin	452	0	0	0	7	10	11	52	61	8	1	0	0	(17)
(18)		PrecStd	160	9	10	11	24	38	54	96	82	53	29	28	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	17.1	25.0	30.7	34.9	37.2	36.7	35.3	32.7	28.5	22.6	14.1	(19)
(20)			MCWB	4.3	9.9	13.6	18.6	18.9	20.7	24.0	26.1	22.5	17.4	12.8	6.7	(20)
(21)		2%	DB	9.4	14.0	21.9	27.8	31.9	35.2	35.1	33.6	30.7	26.1	19.8	11.3	(21)
(22)			MCWB	3.2	6.9	12.0	16.9	18.8	20.9	25.2	25.9	21.6	16.1	11.5	5.3	(22)
(23)		5%	DB	7.4	12.1	19.7	25.8	30.3	33.8	33.9	32.2	29.2	24.3	17.7	9.4	(23)
(24)			MCWB	2.0	5.6	10.7	15.8	18.2	20.9	25.1	25.1	20.5	15.3	10.2	3.8	(24)
(25)		10%	DB	5.7	10.3	17.4	23.8	28.8	32.5	32.7	31.1	27.7	22.6	15.7	7.7	(25)
(26)			MCWB	0.9	4.4	9.2	14.4	17.6	20.7	24.7	24.6	19.8	14.7	9.3	3.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	10.4	15.7	20.9	22.2	25.7	28.1	27.9	25.6	19.8	14.6	8.2	(27)
(28)			MCDWB	9.8	16.9	23.0	27.5	28.4	32.0	33.6	32.8	29.6	24.0	20.2	11.9	(28)
(29)		2%	WB	3.7	7.8	13.1	18.5	21.1	24.3	27.2	27.1	23.5	18.5	12.7	6.2	(29)
(30)			MCDWB	8.2	12.9	20.4	25.8	28.1	29.9	32.4	31.7	28.1	22.7	17.1	9.6	(30)
(31)		5%	WB	2.5	6.3	11.4	17.0	20.1	23.3	26.7	26.5	22.2	17.3	11.4	4.9	(31)
(32)			MCDWB	6.8	11.1	18.5	24.0	27.0	29.4	31.6	30.8	27.0	22.0	16.3	8.2	(32)
(33)		10%	WB	1.3	4.8	9.7	15.5	19.1	22.4	26.1	25.8	21.2	16.0	10.0	3.6	(33)
(34)			MCDWB	5.0	9.7	16.5	22.4	26.4	29.0	30.9	29.6	25.8	21.0	14.8	6.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.1	7.7	8.6	9.3	9.3	9.1	7.6	7.3	8.3	8.3	7.4	7.0	(35)
(36)			MCDBR	9.0	9.5	10.2	10.4	10.5	10.7	8.9	8.5	9.7	9.7	8.9	8.8	(36)
(37)			MCWBR	5.7	5.8	5.4	5.1	4.5	3.7	3.2	3.2	3.9	4.2	4.7	5.3	(37)
(38)		5% WB	MCDWB	8.0	8.8	9.7	10.3	9.2	8.8	7.6	7.3	7.9	7.9	7.6	7.4	(38)
(39)			MCWBR	5.3	5.9	5.8	5.8	4.9	3.8	3.4	3.1	3.9	4.1	5.0	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.524	0.597	0.613	0.636	0.660	0.796	0.771	0.738	0.674	0.676	0.589	0.529	(40)	
(41)		taud	1.750	1.638	1.631	1.622	1.617	1.431	1.521	1.564	1.632	1.575	1.676	1.763	(41)	
(42)		Ebn at Noon	608	616	663	682	675	588	598	604	611	545	542	568	(42)	
(43)		Edh at Noon	174	219	240	254	259	312	283	267	237	230	186	162	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.31	2.92	4.26	5.06	5.68	5.26	4.66	4.44	3.98	3.20	2.43	2.13	(44)	
(45)		RadStd	0.27	0.43	0.36	0.30	0.41	0.43	0.37	0.49	0.41	0.34	0.37	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.39	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86	N/A

Nomenclature: See separate page